Boxplots

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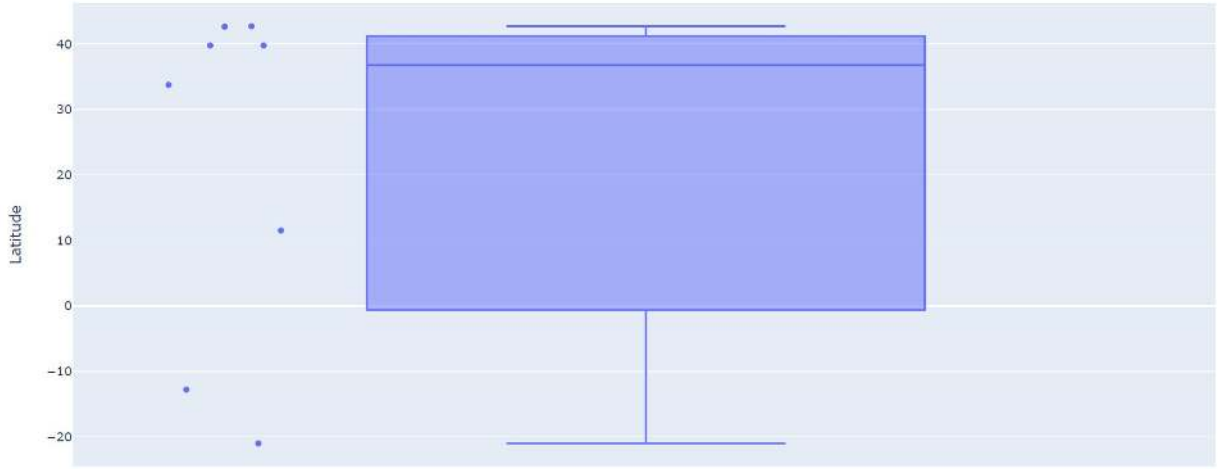
Boxplots are used to show distributions of **numeric** data values. They provide descriptive statistical information at a glance about a group of data's symmetry, skewness, variance, and outliers. Boxplots can display the **five-number summary** of a set of data: the minimum, first quartile, median, third quartile, and maximum. In a box plot, we draw a box from the first quartile to the third quartile. A vertical line goes through the box at the median.

The NLP Suite provides several options for Boxplots.

The Boxplot tool typically graphs each data point alongside a boxplot produced from the data. The result is a side-by-side frame with a scatter plot and a boxplot. However, this scatterplot can be adjusted under the Data Points widgets in the Boxplot parameters section. You will select "All points" most often, but you can select the "Outliers only" to examine the distribution and scope of data points that might be skewing your data. However, if you wish to only generate a boxplot and exclude the scatterplot with all the points, simply select the option for "No points" in the Data points widget.

Suppose that you want a boxplot for the Latitude values you have obtained from running the GIS scripts.

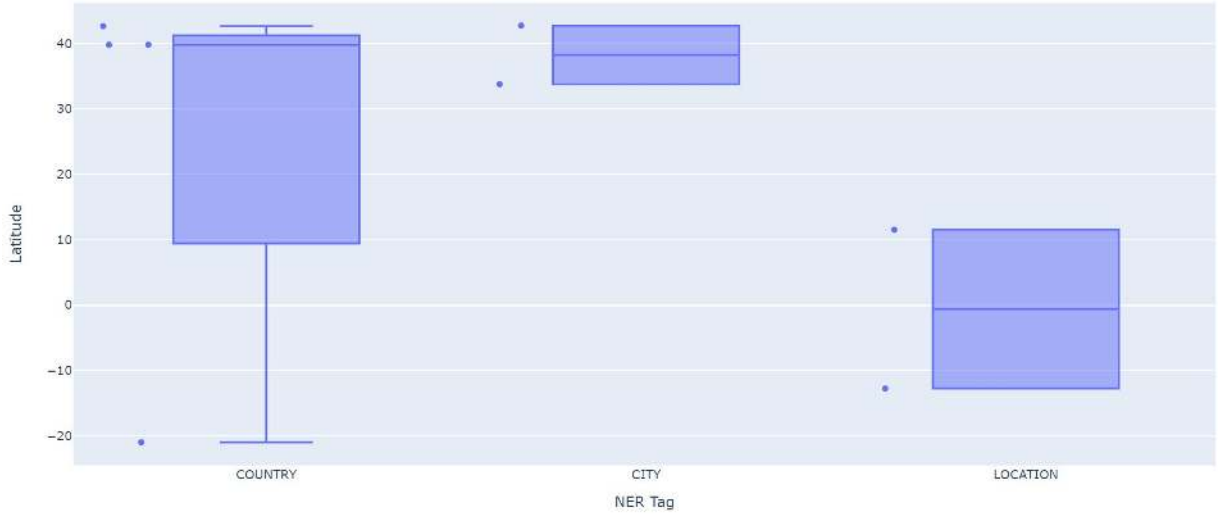


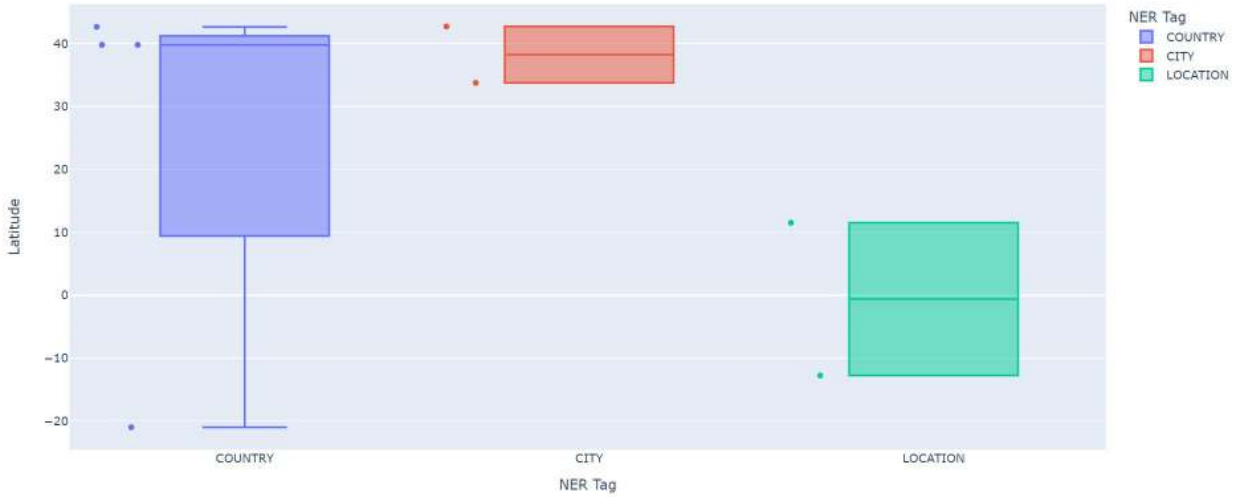


You can also option to split data by category, for instance, Latitude by type of NER location tags (e.g., COUNTRY, CITY).



By selecting “Split data by category,” you can group the boxplots by a selected CATEGORY field. By selecting the first extra CATEGORICAL csv file field, you can add in an X-axis on which the points will be graphed. By selecting a second extra CATEGORICAL csv file field, you can add color to your plots.

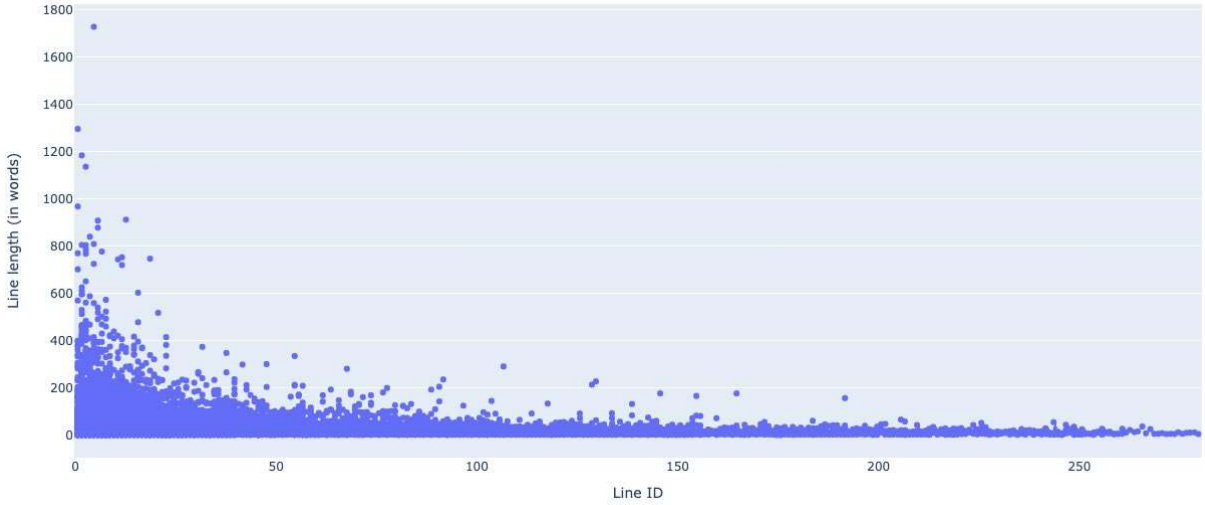




The following example compares line length (in words) in the HIV in Africa Corpus. All data points were graphed, and the data was not split by category.



The user can observe that there is a clear outlier. To identify the source of this outlier, you could split the data by Line ID.


1800
1600
1400
1200
1000
800
600
400
200
0
Line length (in words)
0
50
100
150
200
250
Line ID